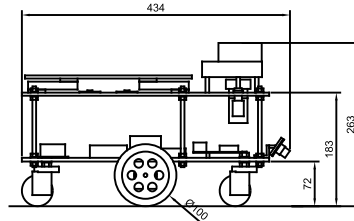
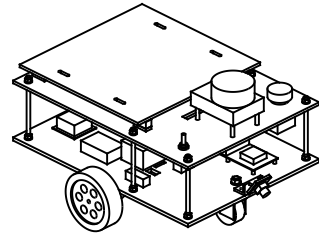


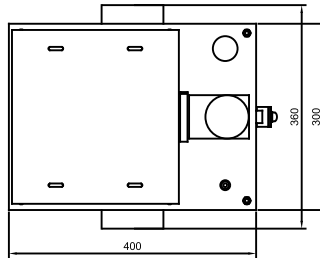
FRONT VIEW



SIDE VIEW (from the right) FRONT -->



ISOMETRIC VIEW

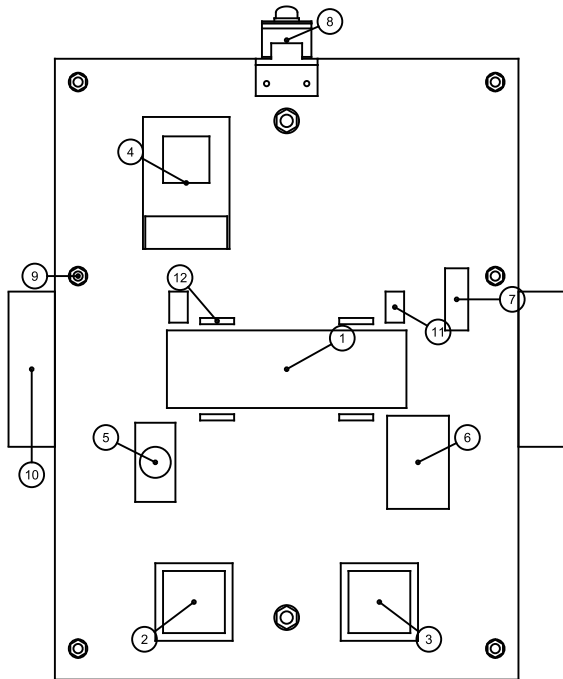


PLAN VIEW FRONT -->

NOTES

1. All dimensions in mm. Weights and loads in kg.
2. Drive: 2 x JGB37-520 12 V 170 RPM gear motors, wheels D100 x 30.
3. Casters: 2 x 50 mm swivel, M8 threaded stem (height set by nuts).
4. Frame: 6 mm plywood + 5 mm acrylic on 6 x M6 threaded rod.
5. Payload: 4 x 10 kg load cells under the cargo plate, max 15 kg.
6. Sensors: RPLIDAR A1M8 (front), USB camera tilted 35 deg down.
7. Sheets: 01 GA, 02 layout, 03-05 plates, 06 printed parts, 07 BOM.

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
GENERAL ARRANGEMENT				CR-V2-001	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:5	1 / 7	mm	1st ANGLE	28/09/2026	



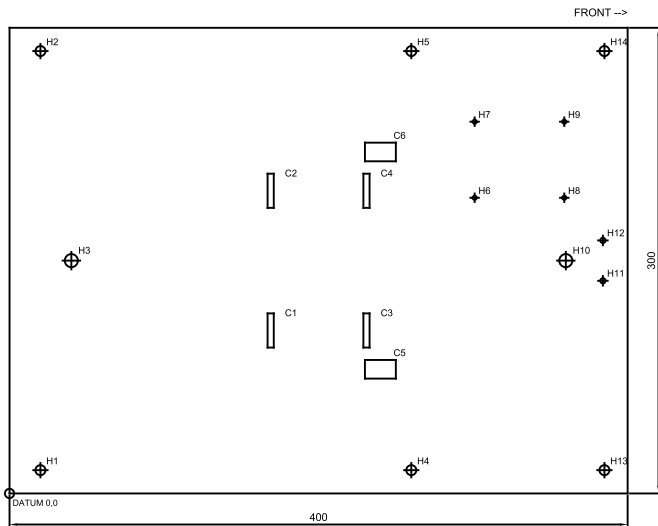
PLAN VIEW - DECK, CARGO PLATE AND LIDAR REMOVED (front of robot at the top)

No.	COMPONENT
1	3S LiPo battery 5000 mAh (lies across, slides out sideways)
2	BTS7960 motor driver - left motor
3	BTS7960 motor driver - right motor
4	Raspberry Pi 5 on M2.5 standoffs
5	Buck converter 12 V -> 5.1 V
6	WAGO power buses (+12 V main / motor / GND)
7	Inline fuse 10 A
8	USB camera on printed mount
9	M6 threaded rod (x6)
10	Drive wheel D100 (motor under the plate)
11	Motor / encoder cable slot (x2)
12	Battery strap slots (x4)

NOTES

Small boards are held with VHB tape; the Pi on M2.5 standoffs.
 Keep 10 cm slack on every wire that goes up to the deck.
 Wiring: see BUILD_GUIDE.md, weeks 2 and 3.

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
LAYOUT - DECK REMOVED				CR-V2-002	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	2 / 7	mm	1st ANGLE	28/09/2026	



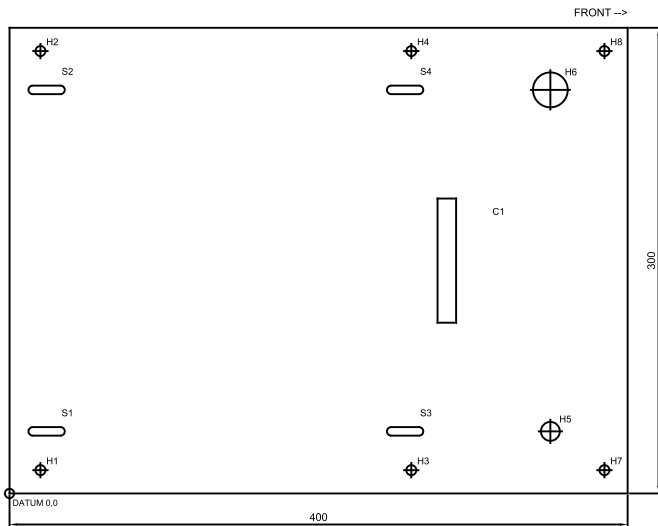
NOTES

1. Material: 6 mm birch plywood (or 5 mm aluminium), 1 off.
2. Laser-cut from cad/output/dxf/bottom_plate.dxf. All dimensions in mm, +0.2.
3. Motor-bracket holes (4 x D4.5) are drilled on assembly - see BUILD_GUIDE week 1.

ID	X	Y	SIZE	FOR
H1	20.0	15.0	D6.5	M6 threaded rod
H2	20.0	285.0	D6.5	M6 threaded rod
H3	40.0	150.0	D8.5	caster stem M8
H4	260.0	15.0	D6.5	M6 threaded rod
H5	260.0	285.0	D6.5	M6 threaded rod
H6	301.0	190.5	D2.7	Raspberry Pi M2.5
H7	301.0	239.5	D2.7	Raspberry Pi M2.5
H8	359.0	190.5	D2.7	Raspberry Pi M2.5
H9	359.0	239.5	D2.7	Raspberry Pi M2.5
H10	360.0	150.0	D8.5	caster stem M8
H11	384.0	137.0	D3.4	camera mount M3
H12	384.0	163.0	D3.4	camera mount M3
H13	385.0	15.0	D6.5	M6 threaded rod
H14	385.0	285.0	D6.5	M6 threaded rod
C1	169.0	105.0	4 x 22	battery strap
C2	169.0	195.0	4 x 22	battery strap
C3	231.0	105.0	4 x 22	battery strap
C4	231.0	195.0	4 x 22	battery strap
C5	240.0	80.0	20 x 12	motor/encoder cables
C6	240.0	220.0	20 x 12	motor/encoder cables

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
BOTTOM PLATE				CR-V2-003	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	3 / 7	mm	1st ANGLE	28/09/2026	



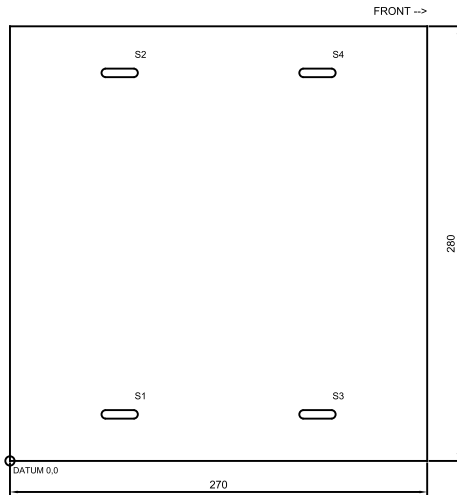
NOTES

1. Material: 5 mm clear acrylic (keep the protective film), 1 off.
2. Laser-cut from cad/output/dxf/top_deck.dxf. All dimensions in mm, +0.2.
3. Lidar holes (4 x D2.7) are drilled on assembly, using the lidar as template.

ID	X	Y	SIZE	FOR
H1	20.0	15.0	D6.5	M6 threaded rod
H2	20.0	285.0	D6.5	M6 threaded rod
H3	260.0	15.0	D6.5	M6 threaded rod
H4	260.0	285.0	D6.5	M6 threaded rod
H5	350.0	40.0	D12.2	master switch
H6	350.0	260.0	D22.5	E-stop 22 mm
H7	385.0	15.0	D6.5	M6 threaded rod
H8	385.0	285.0	D6.5	M6 threaded rod
S1	24.0	40.0	23.5 x 5.5 (R2.75)	load-cell screws
S2	24.0	260.0	23.5 x 5.5 (R2.75)	load-cell screws
S3	256.0	40.0	23.5 x 5.5 (R2.75)	load-cell screws
S4	256.0	260.0	23.5 x 5.5 (R2.75)	load-cell screws
C1	283.0	150.0	12 x 80	wiring pass-through

X, Y = centre of the feature from DATUM 0.0 (H = hole, S = slot, C = cut-out)

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
TOP DECK				CR-V2-004	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	4 / 7	mm	1st ANGLE	28/09/2026	



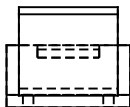
ID	X	Y	SIZE	FOR
S1	71.0	30.0	23.5 x 5.5 (R2.75)	load-cell screws
S2	71.0	250.0	23.5 x 5.5 (R2.75)	load-cell screws
S3	199.0	30.0	23.5 x 5.5 (R2.75)	load-cell screws
S4	199.0	250.0	23.5 x 5.5 (R2.75)	load-cell screws

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

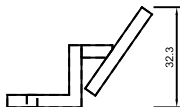
NOTES

1. Material: 5 mm clear acrylic, 1 off. Anti-slip mat on top.
2. Laser-cut from cad/output/dxf/cargo_plate.dxf. All dimensions in mm, +0.2.
3. Rests only on the 4 load cells (slots S1-S4).

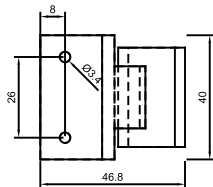
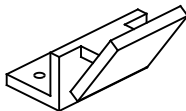
PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
CARGO PLATE				CR-V2-005	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	5 / 7	mm	1st ANGLE	28/09/2026	



FRONT



CAMERA MOUNT - SIDE

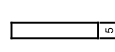
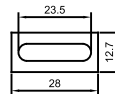


PLAN

Holder face tilted 35 deg down. Drill 2 mm holes to suit the camera board.

NOTES

1. Material PETG, 0.2 mm layers, 4 walls.
2. Camera mount: 1 off, 40 % infill, supports on build plate only.
3. Load-cell spacer: 8 off, 100 % infill, print flat.
4. STL files: cad/output/parts/



LOAD-CELL SPACER (8 off), slot 5.5 wide

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
PRINTED PARTS				CR-V2-006	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:1	6 / 7	mm	1st ANGLE	28/09/2026	

No.	GROUP	DESCRIPTION	QTY	MAKE / BUY
1	Compute	Raspberry Pi 5 8GB + Active Cooler + 64GB A2 microSD (x2)	1	buy
2	Drive	JGB37-520 12V 170 RPM gear motor with Hall encoder (6 wires)	2	buy
3	Drive	Steel mounting bracket for 37 mm gear motor (+ its M3 screws)	2	buy
4	Drive	Wheel 100 mm x 30 mm with 6 mm D-shaft hub	2	buy
5	Drive	Swivel caster 50 mm, M8 threaded stem, >= 20 kg rating	2	buy
6	Drive	BTS7960 (IBT-2) 43 A motor driver	2	buy
7	Sensors	RPLIDAR A1M8 (with USB adapter)	1	buy
8	Sensors	USB (UVC) camera board 32x32 mm, wide-angle M12 lens	1	buy
9	Sensors	Camera mount (camera_mount.stl)	1	3D print PETG
10	Payload	Straight-bar load cell 10 kg, 80x12.7x12.7 mm	4	buy
11	Payload	HX711 module	4	buy
12	Payload	Load-cell spacer (load_cell_spacer.stl)	8	3D print PETG 100 %
13	Payload	Cargo plate 270x280x5 (cargo_plate.dxf)	1	laser-cut acrylic
14	Payload	Anti-slip rubber mat for the cargo plate	1	buy
15	Power	3S LiPo 11.1 V 5000 mAh XT60 + balance charger + LiPo bag + voltage alarm	1	buy
16	Power	XL4015 5 A buck converter + USB-C pigtail	1	buy
17	Power	Inline fuse holder + 10 A fuses, WAGO 221 connectors, XT60 pairs	1	buy
18	Power	Emergency stop 22 mm panel-mount, latching, NC contact block	1	buy
19	Power	Toggle switch 12 mm panel-mount, 12 V DC >= 15 A	1	buy
20	Chassis	Bottom plate 400x300x6 (bottom_plate.dxf)	1	laser-cut 6 mm plywood
21	Chassis	Top deck 400x300x5 (top_deck.dxf)	1	laser-cut 5 mm acrylic
22	Chassis	M6 threaded rod, cut to 130 mm	6	cut from 1 m rod
23	Chassis	M6 nut + M6 washer	24	buy
24	Chassis	M8 nut + M8 washer (casters)	4	buy
25	Hardware	M4x12 screws + nylock nuts (motor brackets), M4/M5x16 (load cells), M3/M2.5 screws + standoffs (Pi, camera), VHB tape, velcro strap, zip ties	1	buy

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
BILL OF MATERIALS				CR-V2-007	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
-	7 / 7	mm	1st ANGLE	28/09/2026	